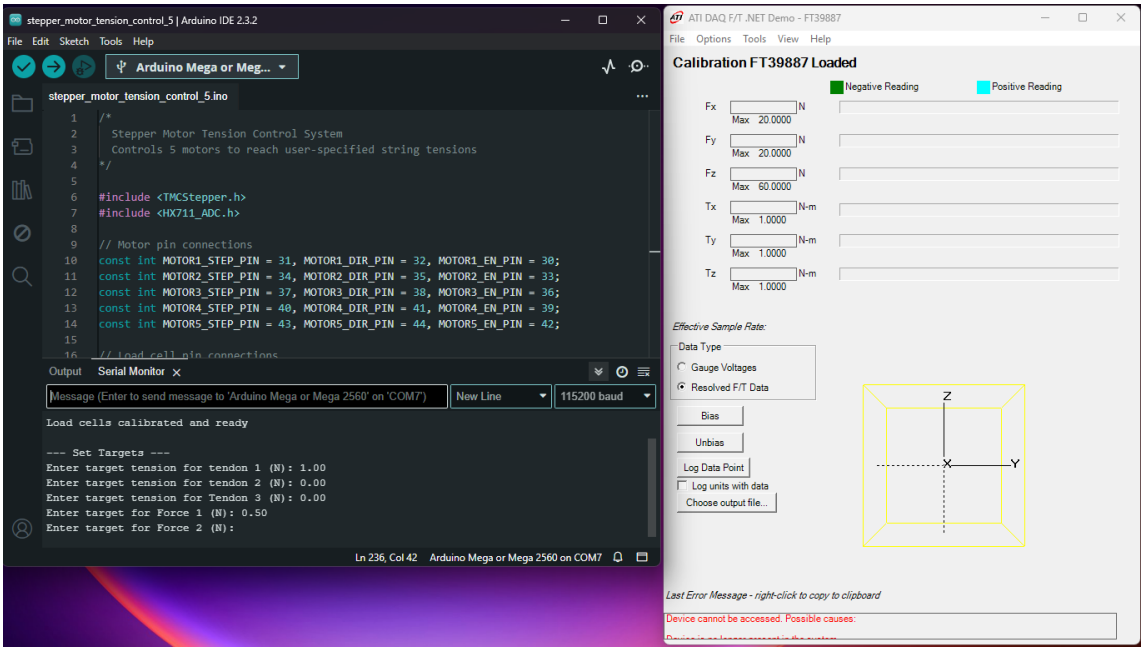
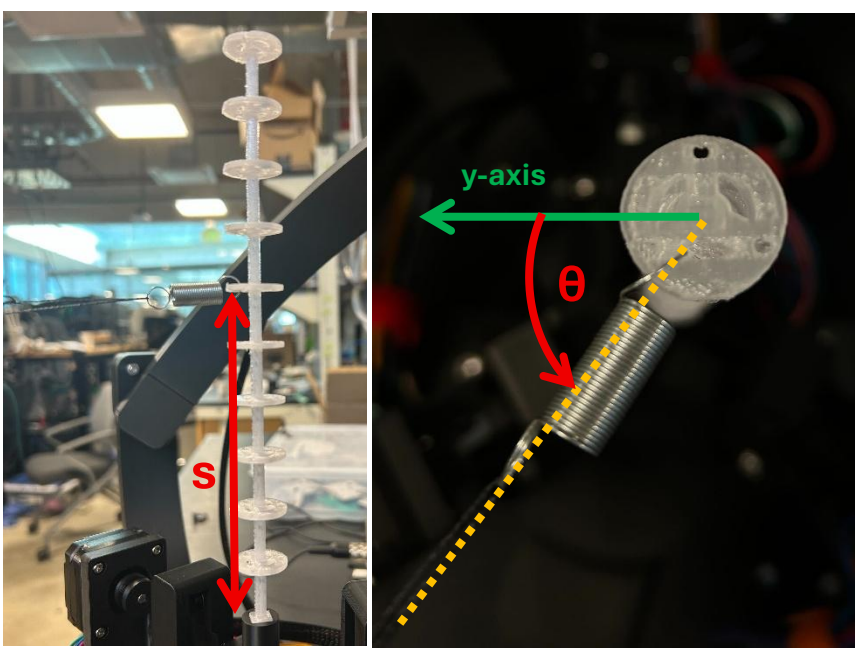
Experimental Procedure

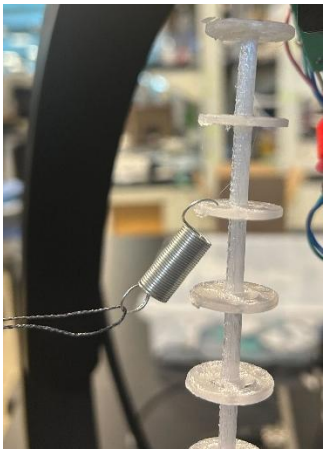
1. **System Initialization:** The ATI force/torque sensor, microcontroller, data logging software, and motor control script were powered on and initialized.



2. **Configuration:** Loading cables were attached to the robot at arc lengths s_1 and s_2. The test rig gimbal arms were pre-positioned so that the loading cables create approximate angles θ_1 and θ_2. Loading cables were attached to the robot at arc lengths s_1 and s_1.



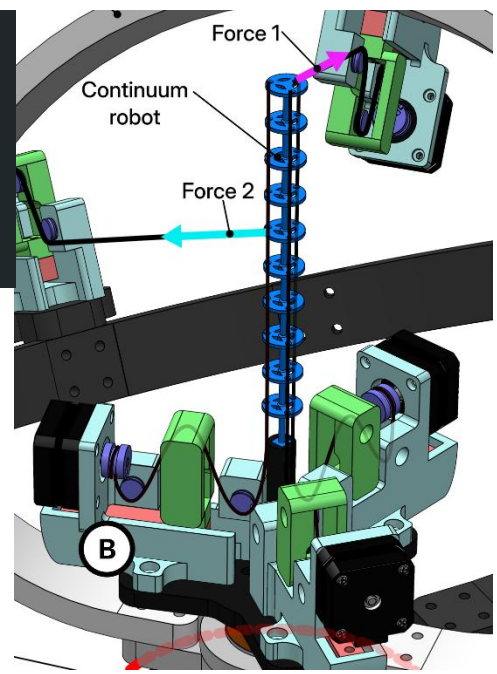
3. **Calibration:** With all cables slack to prevent residual loading, the ATI sensor was tared to zero.



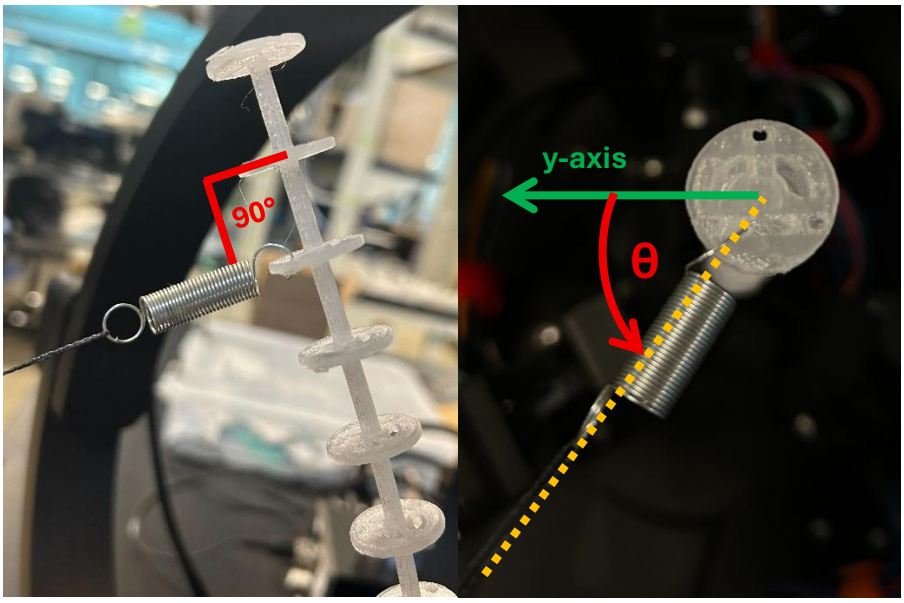
4. **Actuation:** Target tendon tensions τ_1, τ_2, τ_3 and external loads F_1 and F_2 were entered in the motor control script, triggering the winches to tension the cables automatically.

```
Load cells calibrated and ready

--- Set Targets ---
Enter target tension for tendon 1 (N): 1.00
Enter target tension for tendon 2 (N): 0.00
Enter target tension for Tendon 3 (N): 0.00
Enter target for Force 1 (N): 0.50
Enter target for Force 2 (N):
```



5. **Geometric Alignment:** Upon reaching steady-state equilibrium, the gimbal arms were manually adjusted to correct deformation-induced shifts. This ensured the loading cables remained perpendicular to the backbone at their points of contact and aligned with target angles θ_1 and θ_2 in the loaded state.



6. **Re-tensioning:** To mitigate hysteresis induced by these manual adjustments, the motors were relaxed, and the calibration and actuation steps were repeated.
7. **Data Acquisition:** Once the robot reached steady-state equilibrium and cable alignment was visually confirmed, the proximal wrench was recorded.
8. **Iteration:** This procedure was repeated for all test configurations.